

Model 3 → Semantic / Classified 3D View

What the user sees →

- **3D semantic representation of the reconstructed survey area.**
- Actual terrain shape is preserved; the scene is **not flattened**.
- Buildings, roads, terrain, vegetation, trees, forest, water and other structures are separated into semantic classes.
- **Vegetation is further differentiated by observed visual color**, so seasonal or visually distinct vegetation can be represented separately.
- Example:
 - Green Trees
 - Yellow Trees
 - Orange Trees
 - Red/Brown Trees
 - Green Crops
 - Yellow/Dry Crops
 - Yellow Flowers
 - Red Flowers
 - Purple/Blue Flowers
 - White Flowers
- A flower field with a consistent observed color can appear as a distinct semantic region.
- Historical/important structures use the defined historical color; normal buildings use the normal building color.
- Confidence/uncertain regions can be visually distinguished.

What the user can do →

- Rotate, pan and zoom around the 3D semantic scene.
- **Show / Hide / Highlight / Isolate** semantic classes.
- Filter by:
 - Buildings
 - Historical/Important Buildings
 - Roads
 - Trees
 - Forest
 - Crops/Low Vegetation
 - Flowers/Flower Fields
 - Water
 - Other Structures
- Filter vegetation by **observed color**.
- Change rendering between:

- **Semantic**
- **Original RGB / Realistic**
- **Elevation**
- **Confidence**
- Select individual objects or regions.
- Measure and inspect selected geometry.

What happens when they click / select something →

- **Building** → Object ID, class, height, footprint, dimensions, base/roof elevation.
- **Terrain** → elevation, coordinates and terrain information.
- **Road** → road class and associated geometry.
- **Tree/vegetation** → vegetation class, observed color class, confidence and location.
- **Flower field** → flower/vegetation class, observed color, area and location.
- Selected objects/regions are highlighted.
- The selection retains its **Point ID / Object ID / geographic identity** for synchronization with the other views.

Measurement & Inspection workflow →

- Measurements use the **georeferenced metric 3D geometry**.
- User can inspect:
 - Object dimensions
 - Building height
 - Footprint/area
 - Elevation
 - Distance
 - Selected-region area
- For an area selected in Cesium, the corresponding **terrain and structures are highlighted here**.
- Semantic selection can be passed back to Cesium and Point Cloud using the same spatial/object identity.

Layer / Filter controls →

- **Semantic Classes**
- **Vegetation Color Classes**
- **Original RGB**
- **Elevation**
- **Confidence**
- **Historical / Important Structures**
- **Buildings**
- **Roads**
- **Trees**

- **Forest**
- **Crops / Low Vegetation**
- **Flowers / Flower Fields**
- **Water**
- **Other Structures**
- Show / Hide / Highlight / Isolate / Opacity controls.

How switching between modes works →

- Switching between **Semantic, Point Cloud and Cesium** does not run ML again.
- All semantic classifications, object associations and geometry are prepared during the reconstruction job.
- The frontend only loads/streams the required precomputed data.
- Switching modes should therefore be immediate/seamless apart from asset loading.

How the three modes stay synchronized →

Use the common identity:

Point ID → Object ID → Geographic XYZ → Semantic Object → Cesium Geometry

For example, if the user selects a **yellow tree region**:

- Point Cloud → corresponding points are highlighted.
- Cesium → corresponding geographic location/geometry is shown.
- Semantic View → the complete corresponding tree/object/region is highlighted.

If the user selects an **area in Cesium**:

- Corresponding Point Cloud points are highlighted.
- Corresponding terrain and structures are highlighted in Semantic View.

Is this possible? Best approach →

Yes. The best approach is to separate **semantic identity** from **visual color**.

Use:

Object type/classification + observed color classification

rather than treating color itself as the object class.

For example:

Tree + Yellow → Yellow Tree

Flower Field + Purple → Purple Flower Field

Crop + Yellow → Yellow Crop

The original RGB should also be preserved, so semantic colors do not destroy the actual appearance.

Accuracy priority:

The ≤ 1 m target applies primarily to **geometric/georeferencing accuracy**, not semantic color classification. ≤ 1 m can be a validated target with good GPS/RTK/PPK, camera calibration, synchronization, metric reconstruction and QA; it cannot be guaranteed from ordinary GPS + RGB video alone.

Processing priority:

<15 minutes for a 10-minute video is an engineering target and can be approached using adaptive keyframes, GPU processing, AMB3R-SfM, efficient fusion and selective refinement. It must be benchmarked on representative hardware/data.

Requirement in model →

Phase 2 → Data Preparation

- Frame quality assessment.
- Color-preserving image normalization.
- Adaptive keyframe selection.
- Dynamic object masking.

Phase 3 → 3D Reconstruction

- AMB3R-SfM.
- Metric depth.
- Camera poses.
- Pointmap generation/fusion.
- Reconstruction confidence.

Phase 4 → Scene Understanding

- Multi-class aerial semantic segmentation.
- 2D → 3D semantic projection.
- Persistent Object IDs.
- **Vegetation sub-classification.**
- **Vegetation/flower color analysis.**
- **Observed color classification.**
- Color confidence.
- Dynamic vegetation/object handling where required.

Recommended semantic structure:

Vegetation → Type → Observed Color

Examples:

- Tree → Green
- Tree → Yellow
- Tree → Orange
- Crop → Green
- Crop → Yellow
- Flower → Purple
- Flower → Red
- Flower → White

Important: the model should **not claim species or seasonal state from color alone**. Color is an observed visual attribute.

Phase 5 → 3D Object & Terrain

- Project color classes into 3D.
- Group consistent vegetation/flower regions.
- Generate semantic 3D objects/regions.
- Preserve RGB + semantic label + color label.

Phase 6 → Refinement & Georeferencing

- Georeference semantic geometry.
- Confidence filtering.
- Remove inconsistent semantic/color assignments.
- Validate spatial accuracy.

Phase 7 → Output

- Classified 3D geometry.
- Semantic labels.
- Observed color labels.
- RGB information.
- Object IDs.
- Point IDs.
- Confidence values.
- Metadata.

Backend & model flow during loading →

Video → Preprocessing → Adaptive Keyframes → AMB3R-SfM → Metric Depth + Camera Poses + Pointmaps → Point Cloud Fusion → GPS/Telemetry Alignment → Georeferencing

Then:

Georeferenced Point Cloud → Semantic Segmentation → 2D→3D Projection → Object/Region Creation → Vegetation Type Classification → Color Analysis → Color Subclassification → Confidence Filtering

Example:

Pixels → Tree Mask → Tree Object → RGB/Color Analysis → Yellow Tree → 3D Yellow Tree Object

Then backend prepares and stores:

- Point ID
- Object ID
- XYZ
- Latitude/Longitude
- Elevation
- Terrain elevation
- RGB
- Semantic class
- **Observed color class**
- Semantic confidence
- Color-class confidence
- Spatial index
- Point ↔ Object mapping

Finally, it prepares the three visualization assets:

Point Cloud → LAS/LAZ/PLY

Semantic → Classified 3D geometry + Object IDs + semantic/color attributes

Cesium → Georeferenced GLB/3D Tiles + Object IDs

All of this is completed **during the reconstruction/loading job**, so the Semantic View does not need to run segmentation or color analysis when the user switches to it.

